# Uddhav P. Gautam (Ph.D. Candidate)

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**Software Engineer, Tech Lead, and Architect** — shipping production systems at scale

## Summary

A decade building and leading production software, now with systems-and-security research across the stack — from the Linux kernel and embedded devices to the software supply chain and AI-assisted development. Positioned for the senior technical-leadership and research roles that follow: principal architect, engineering director, or research scientist.

## Research Interests

- **AI Safety**: verification of AI-generated code; resilience and power safety of AI data centers
- **Security**: embedded systems; software supply chains; mobile and wearable forensics; operational technology
- **Systems**: eBPF; Linux kernel; Android OS/internals; on-the-fly over-the-air updates for microcontrollers
- **Software Engineering**: release assurance; verification and validation; testing of diagnostic APIs

## Professional Experience

**Android Technical Lead & Mobile Architect, Perficient** *Jun 2021 – May 2023*

- Architected infrastructure to migrate 30M+ TD Ameritrade users (Android, iOS, web) to Charles Schwab
- Led an 8-engineer Android team on a regulated codebase; set engineering and documentation standards
- Optimized a shared C++ core and a mobile performance pipeline, helping cut production crash rate 40%
- Migrated builds to Kotlin DSL, added static analysis, and restructured CI/CD for multi-flavor releases

**Senior Android Developer, Viper Design LLC** *Mar 2020 – Feb 2021*

- Built the IoT stack for Shark robot vacuums, from embedded firmware to cross-platform mobile apps
- Re-architected the Android app around modular design and dependency injection to improve testability
- Improved stability through performance tuning, code shrinking, crash analytics, and automated testing

**U.S. Army Reserve** *Apr 2019 – Apr 2023*

- Served 4 years; scored in the $96^{\text{th}}$ percentile on the Armed Forces Classification Test

***Prior roles**: 3 years as Android Developer; before that, 4 years as System/Network Administrator.*

## Research & Teaching Experience

### Research Leadership

**Principal Investigator**, Meta Project Aria Gen 2 Research Partnership *2026 – Present*

- Lead the lab's Aria Gen 2 wearable-security research program and set its research directions
- Serve as primary liaison with Meta Reality Labs Research
- Collaborate on all Aria-based lab projects and publications

## Research Highlights

**ZeroDown**: on-the-fly eBPF security-policy updates for microcontrollers *In preparation, IEEE IoT-J*

- Idea: the device's owner writes its security policy as one versioned YAML file — the rules, exceptions for particular peers, and the name of the eBPF program that decides each event — so a change means editing the file, not the firmware
- Built the system that installs a new policy file on a Bluetooth Low Energy (BLE) device and activates it without a reboot, instead of waiting months for a firmware update and a restart
- In hardware tests, all 72 updates applied without a reset, taking a median 2.8 s; in a separate 60-update test with a peer connected, the connection never dropped and the device's notifications to the peer paused for a median 2.5 s
- Designed the update process: authenticated delivery, crash-consistent storage in two flash slots, and a confirmation step — the device keeps a new policy only if the owner reconnects and confirms it, and reverts it otherwise
- Protected against bad updates: in all 7 tests with a corrupted or interrupted update, the existing policy stayed in place; and when a new policy locked the owner out, resetting the device restored the previous one in all 120 trials
- Hardened the eBPF interpreter that runs the rules (step limit, memory checks, divide-by-zero fix): no crash in 146.9M host-side fuzzing executions; deciding one event takes about 0.3 ms

**Android 17 JobScheduler**: its reports of why jobs wait can be wrong while its scheduling is right
Lost Boundaries (in preparation, IEEE TMC); Correct Decisions (under review, ACM HotMobile '27)

- Found that Android 17's report can show a job waiting longer than it has existed, and show a system policy (App Standby) holding the job longer than the app's own requested delay — pointing a developer at the wrong cause (30 of 30 qualifying emulator attempts)
- Showed the reported wait shrinking by over 8 s while the job kept waiting — a decrease no correct count can produce (2 of 2 emulator boots)
- Traced the cause: two wait conditions reported under one reason share one start time; the second to start waiting overwrites it, and the first to finish deletes it — so later waits are measured from the wrong moment, even from the job's creation
- Proved that two different wait histories can leave identical stored data, so no change to the calculation alone can report both correctly
- Explained why the error goes unnoticed: scheduling never reads the reported durations, so it stays correct, and none of the 7 upstream tests for this report checks the duration of a shared reason
- Proposed a 4-part test format (property, history, observations, assertion) to catch such errors
- Built a one-commit fix: on both repaired emulator images, neither the ranking reversal nor the reported decrease reappeared

**AIE Verification**: approving AI-written code and software releases *In preparation, ACM TOSEM*

- Showed that AI-written tests can pass wrong code: in 25 constructed LLM coding tasks (Codex CLI), all passed their own tests, including 12 labeled flawed for following a prompt's wrong assumption
- Wrote checks from the real requirement: they rejected all 12 flawed results and accepted the other 13
- Defined 3 conditions for trusting an approval: it checks the real requirement, its inputs are controlled outside the build pipeline, and it applies to the exact file delivered
- Built a release check in an Android app's release CI/CD pipeline: it rejected all 9 bad test inputs and accepted all 4 good ones

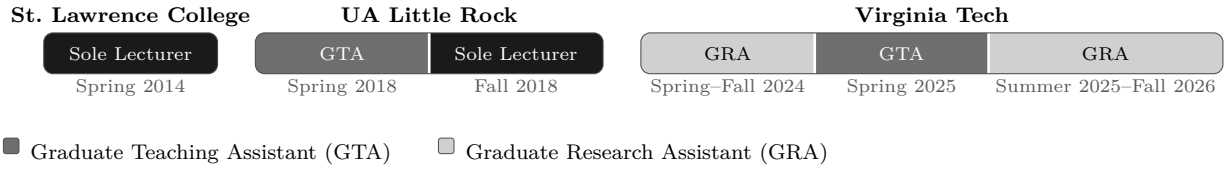
**eBPF & HELOT**: Linux kernel tracing and power-grid forensics

- Co-authored a published method that removes the slowdown eBPF tracing imposes on untraced programs (ACM SIGCOMM eBPF workshop 2024)

- Co-authored HELOT, a DOE-funded prototype that collects forensic evidence from power-substation devices without shutting them down (IEEE Trans. Smart Grid '23)
- Moved the pbtools message parser into a Linux kernel module; prototyped a message-filtering firewall

### Teaching & Research Appointments

- **Sole Lecturer** — Discrete Mathematics; Android Apps Development, upper-division
- **Graduate Teaching Assistant** — Operating Systems and Microprocessor Programming; Java




## Software

Founder and sole developer, RGR Innovate LLC (2024 – Present): 3 free, ad-free apps on Google Play.

- **Earthquakes by RGR**: real-time USGS earthquake map with location search and magnitude filters
- **SmartConvert**: offline document scanner, image/unit/color converter, and ARCore room measurement
- **Passport & ID Photo Studio**: print-ready passport, visa, and ID photos for 155+ countries, with on-device face detection, background removal, and compliance checks

## Technical Expertise

**Systems & Kernel**
- eBPF (TC, XDP, tracing, kfuncs, uBPF)
- Linux kernel, Android OS/internals
- C/C++, Rust, Python

**Embedded & IoT Security**
- BLE protocols and security (SMP, GATT, L2CAP)
- Zephyr RTOS, Nordic nRF52840 (Arm Cortex-M4)
- Authenticated over-the-air policy updates
- Runtime policy enforcement

**Security & Assurance**
- Software supply-chain security
- Reproducible builds, SBOM, SARIF
- Trivy, OWASP Dependency-Check, MVT, fuzzing
- Mobile/wearable forensics (UFED, Project Aria)
- SCADA/OT security (DNP3, Modbus, IEC 61850)

**Software & Infrastructure**
- Kotlin, Java, Android SDK/NDK
- DevSecOps, Docker, Kubernetes, Terraform, KVM
- AWS, GCP, Firebase
- Elasticsearch, Logstash, Filebeat, GRR

## Education

- **Ph.D. Computer Engineering, Virginia Tech** (GPA: 4.0/4.0) *2024 – Present*
  Advisors: Haining Wang, Randy Marchany
- **M.S. Computer Science, UA Little Rock** (GPA: 3.58/4.0) *2015 – 2017*
- **B.S. Computer Science, Tribhuvan University, Nepal** (GPA: 3.91/4.0) *2009 – 2013*

## Honors & Awards

- **UA Little Rock** — 1st Place: Complete Societal Award, Societal Impact Award, Best Graduate Project Award; 2nd Place: Personal Achievement Award & TechLaunch Entrepreneurial Team Projects Competition
- **Tribhuvan University** — Top-5 Final-Year Project Award, Nepal Telecommunications Authority

## Publications

### Peer-Reviewed Publications

- Milo Craun, Khizar Hussain, **Uddhav P. Gautam**, et al. "Eliminating eBPF Tracing Overhead on Untraced Processes." *ACM SIGCOMM Workshop on eBPF and Kernel Extensions*, 2024. DOI: 10.1145/3672197.3673431
- Syed Akailvi, **Uddhav P. Gautam**, et al. "HELOT–Hunting Evil Life in Operational Technology." *IEEE Transactions on Smart Grid*, vol. 14, no. 4, pp. 3058–3071, Jul. 2023. DOI: 10.1109/TSG.2022.3222261

### Manuscripts Under Review

- **Uddhav P. Gautam** et al. "Correct Decisions, Incorrect Diagnostic Reports." *ACM HotMobile 2027.* Artifact
- Sahil Dudani, **Uddhav P. Gautam**, et al. "FAIR-ARIA: Forensic Activity Indexing for Review Using Sensor-First Analysis of Smart-Glasses Artifacts." *ACM HotMobile.*

### Manuscripts in Preparation

- **Uddhav P. Gautam** et al. "ZeroDown: Authenticated, Owner-Confirmed In-Place Updates of eBPF Security Policies for Microcontroller-Class IoT Devices." *Targeting IEEE Internet of Things Journal (IoT-J).*
- **Uddhav P. Gautam** et al. "Breaking Self-Referential Trust with AIE Verification in AI-Assisted Development and Release Assurance." Artifact. *Targeting ACM Transactions on Software Engineering and Methodology (TOSEM).*
- **Uddhav P. Gautam** et al. "Lost Boundaries: Label-Keyed Boundary Collapse and Exact Repair in Android JobScheduler." *Targeting IEEE Transactions on Mobile Computing (TMC).*
- Sahil Dudani, **Uddhav P. Gautam**, et al. "SPADE: Spyware and Parental-Control Artifact Discovery and Examination on Android Devices." *Targeting Forensic Science International: Digital Investigation.*

## Certifications

Software Architecture, CI/CD & DevOps, Deep Learning (NVIDIA), Java/Kotlin Design Patterns, React Native, Scrum Master, and Microsoft certifications (full list on LinkedIn).

## References

- **Haining Wang**, Professor, Dept. of Electrical & Computer Engineering, Virginia Tech — Ph.D. advisor; co-author on all my Virginia Tech papers. hnw@vt.edu, (757) 813-4390
- **Randy Marchany**, Adjunct Professor, Dept. of Electrical & Computer Engineering, Virginia Tech (former University CISO) — Ph.D. co-advisor; co-author on all my Virginia Tech papers. marchany@vt.edu, (540) 250-7681
- **Brad Tilley**, Director, Security Architecture & Red Team, Virginia Tech — supervised my Mobile DevSecOps pipeline work over two summers; co-author on two of my papers. brad@vt.edu, (540) 231-3133